Project Assignment

2107541 Machine Learning for Environmental Application

Project 1: Individual Project (20%)



Machine Learning in Environmental Applications

Assignment Brief

Select one machine learning application, model, or framework related to an environmental problem and explain how machine learning is being used to address the problem.

Option 1 — Research Paper

- Air pollution prediction
- Water quality prediction
- Flood prediction
- Water treatment plant operation
- Climate prediction
- Energy management system
- Business and Sustainability

Option 2 — Real-World Case

- AI for air-quality forecasting

- ML for flood prediction

- AI-based water quality monitoring

Option 3 — Machine Learning Model / Framework

- Random Forest
- XGBoost
- CatBoost
- LightGBM
- Explainable ML
- Physics-Informed ML
- Hybrid ML models

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Machine Learning in Environmental Applications

Format: 5-7 minutes of recorded video (due date 11 Nov 2026)

1. Environmental Problem
2. Why Machine Learning?
3. Machine Learning Model / Framework
4. Real Application / Research Findings
5. Environmental Meaning / Interpretation
6. Room for Improvement (Your Opinion)



Submit both Presentation slide and Recorded video

Project 1: Individual Project (20%)

Machine Learning in Environmental Applications

Example

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Machine learning application in building energy consumption prediction: A comprehensive review

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ABSTRACT

Energy, the lifeblood of modern society, has garnered increased attention toward managing building energy consumption over the past decade. The proliferation of available data has opened new avenues for leveraging machine learning to predict building energy usage. Despite the extensive literature in this domain, there is a lack of systematic reviews that provide a comprehensive overview of machine learning applications in predicting building energy consumption. This research aims to: (1) review the diverse applications of machine learning in forecasting building energy consumption; (2) summarize recent advancements in machine learning for enhancing building energy efficiency; and (3) identify current research gaps while proposing future trends. Initially, 431 relevant articles published between 2012 and 2023 were examined using bibliometric analysis, leading to the identification of 16 research keywords and 9 clusters. Subsequently, content analysis was employed to assess building types, energy sources, and temporal granularity. Finally, existing research gaps were identified, and six future research directions were proposed, including (1) integration of multi-source heterogeneous data; (2) development of model transfer frameworks across different buildings; (3) increased focus within the computer science community; (4) interdisciplinary collaboration and standardization; (5) real-time monitoring and forecasting of energy consumption; and (6) data security and privacy protection. This research not only highlights prevailing research gaps but also outlines future trajectories, providing valuable insights for researchers and practitioners navigating this dynamic field.

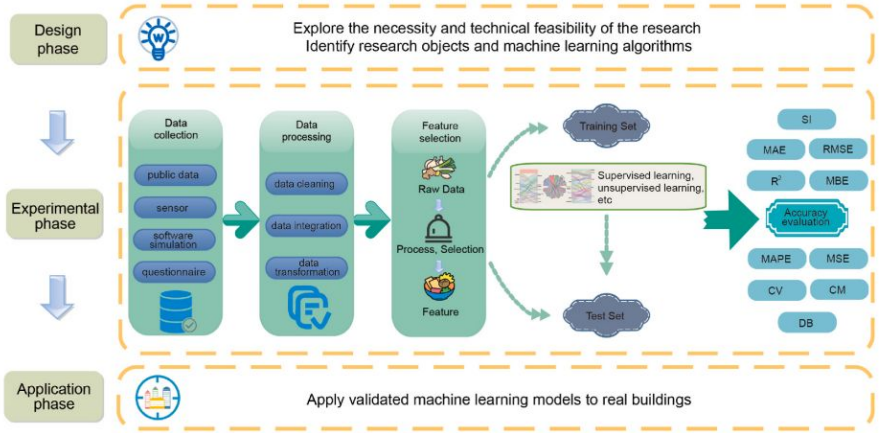


Fig. 6. Experimental framework of building energy prediction.

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Machine Learning in Environmental Applications

Example

sciendo *Economics and Business Review* *Vol. 11 (3), 2025: 91–117*
https://www.ebr.edu.pl *https://doi.org/10.18559/ebr.2025.3.1895*

Clustering S&P 500 companies by machine learning for sustainable decision-making

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Abstract

This study examines the Environmental, Social, and Governance (ESG) performance of S&P 500 companies using three clustering algorithms: *K*-Means, Gaussian Mixture Model, and Agglomerative Clustering. ESG scores from leading data providers are analysed to uncover sectoral patterns and performance trends. The findings indicate that technology and healthcare firms achieve the highest ESG scores, particularly in the governance and social dimensions, while the industrial and energy sectors face the greatest environmental challenges. Among the methods compared, *K*-Means demonstrates superior clustering performance by forming compact and well-separated ESG groups. These results offer a robust foundation for sector-specific ESG benchmarking, supporting investors and policymakers in identifying sustainability leaders, assessing risk, and targeting areas for improvement.

Keywords

- sustainability
- clustering algorithms
- machine learning

JEL codes: Q01, Q56, Q57

Article received 2 December 2024, accepted 1 August 2025.

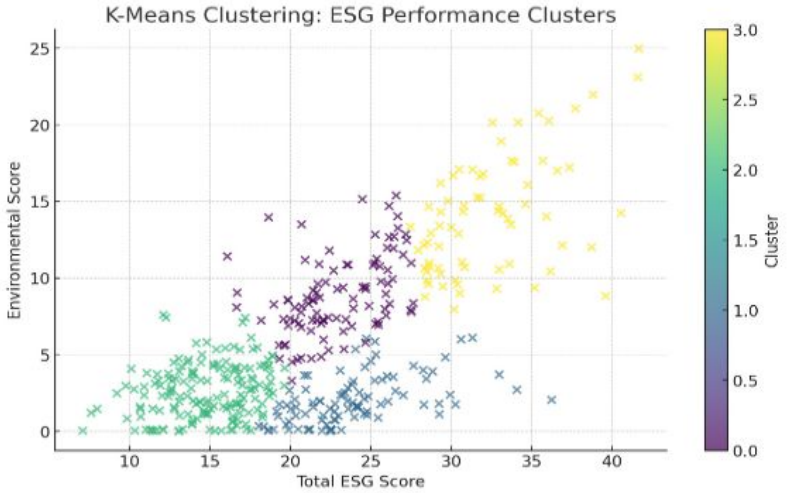


Figure 3. Scatter plot of total ESG score vs. environmental score by *K*-Means clustering

Source: own calculations based on LSEG data.

Project 2: Group Project (20%)

Environmental Machine Learning Project

Group size: 1–3 students
Deliverables: GitHub Repository + Presentation (18 Nov 2026)

Students will develop a complete machine learning workflow using an environmental dataset.

1. Find an Environmental Dataset
2. Define the Problem
3. Data Preparation / Preprocessing
4. Feature Engineering (if any)
5. Build Multiple ML Models (at least 5 for comparison)
6. Hyperparameter Tuning
7. Model Evaluation
8. Feature Importance (Explainable ML)
9. Environmental Interpretation



Project 2: Group Project (20%)

Environmental Machine Learning Project

1. GitHub Repository



```
Environmental-ML-Project/
│
├── README.md
├── data/
├── notebooks/
│   └── ML_analysis.ipynb
├── figures/
└── requirements.txt
```

README must include

- Project title
- Environmental problem
- Dataset source
- Research questions
- Methods
- Main results
- Best model
- Feature importance
- Environmental interpretation

2. Presentation: 10 minutes/group (18 Nov 2026)